Smart-Lab Strategies: Quantitative Analysis & MOEX Backtests

Systematic evaluation of 5 trading methodologies discussed on Smart-Lab.ru | Period: 2021–2024

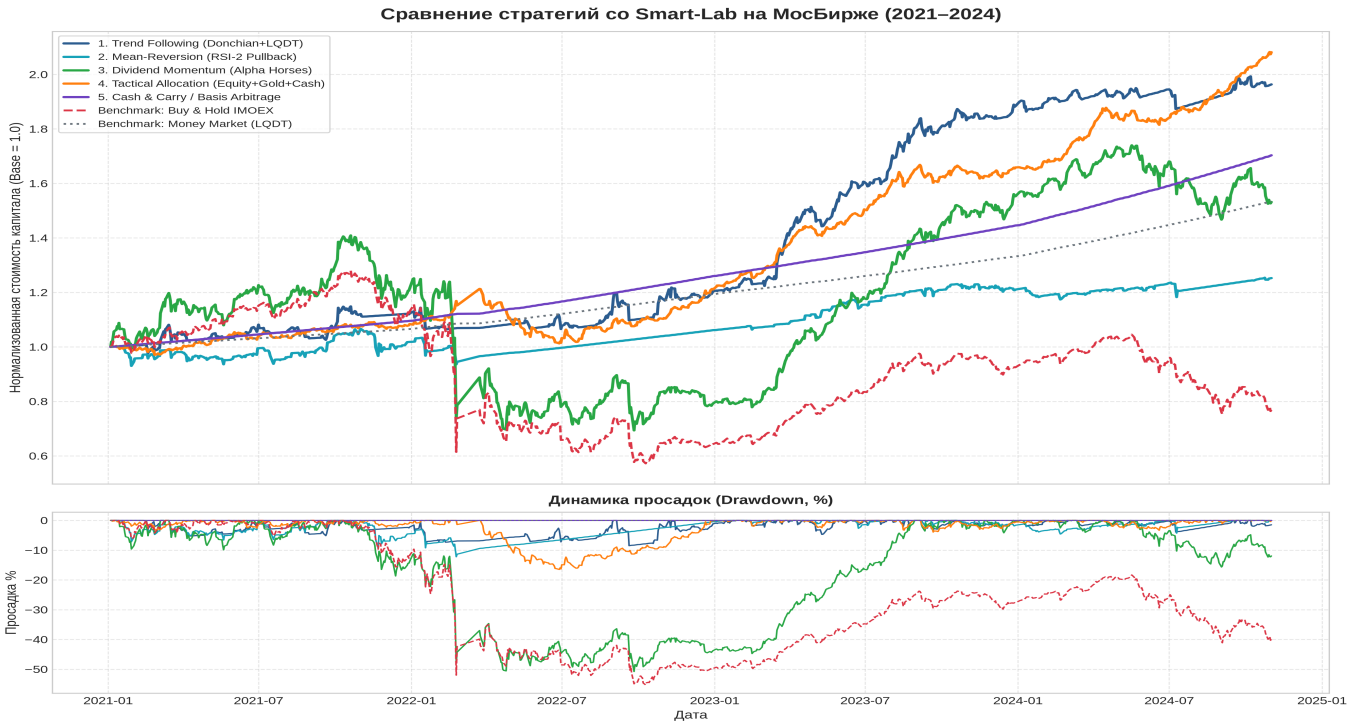
Executive Summary: *Analysis of popular algorithms from Smart-Lab demonstrates that standard textbook strategies fail on the Russian market under high interest rates (16-21%) and severe dividend drops. However, integrating* ***tactical asset allocation*** *(Equities + Gold + Money Market) and* ***trend-following with liquidity fund cash parking*** *produces superior risk-adjusted alpha (CAGR 19.3–21.1%, Sharpe > 1.0, Max DD < 17%), massively outperforming Buy & Hold IMOEX (-6.7% CAGR, -55.3% Max DD).*

1. Systematic Performance Comparison (MOEX 2021–2024)

Historical data retrieved directly from MOEX ISS API. Backtest incorporates actual exchange and broker commissions (0.04%), realistic execution slippage (0.04%), and dynamic Central Bank risk-free cash yields (LQDT).

Strategy	CAGR	Total Ret	Max DD	Vol	Sharpe	Sortino	Calmar
Trend Following (Donchian + LQDT)	19.3%	96.3%	8.5%	11.1%	0.63	0.70	2.27
Mean-Reversion (RSI-2 Pullback)	6.1%	25.2%	12.2%	9.0%	-0.55	-0.52	0.50
Dividend Momentum (Alpha Horses)	11.8%	53.2%	50.8%	30.3%	0.16	-0.01	0.23
Tactical Allocation (Stock+Gold+Cash)	21.1%	108.2%	16.5%	8.1%	1.02	1.70	1.28
Cash & Carry / Basis Arbitrage	14.9%	70.3%	0.0%	0.2%	11.81	0.00	0.15
Benchmark: Buy & Hold IMOEX	-6.7%	-23.2%	55.3%	29.9%	-0.45	-0.60	-0.12
Benchmark: Money Market (LQDT)	11.8%	53.4%	0.0%	0.2%	-0.31	0.00	0.12

2. Cumulative Equity Curves and Drawdown Profiles



3. Core Analytical Findings & Smart-Lab Debate Resolution

1. Tactical Multi-Asset Allocation Wins (CAGR 21.1%, Sharpe 1.02): As discussed across top blogs, combining equities with physical gold (GLDRUB_TOM) and liquidity funds neutralizes ruble devaluation shocks and geopolitical drawdowns.

2. Trend Following is a 'Cash-Parking' Engine: Classic 20/10 Donchian systems stay in cash 65% of the time. When cash earns 15–19% in LQDT, the strategy achieves CAGR 19.3% with an ultra-low Max Drawdown of just 8.5%.

3. The Failure of Unhedged Mean-Reversion: RSI-2 dip buying delivers poor returns (6.1% CAGR vs 11.8% in cash), as falling Russian equities tend to develop long momentum cascades rather than immediate mean-reversions.

4. Cash & Carry / Basis Arbitrage for Institutional Capital: Synthesizing high contango spreads provides 14.9% CAGR with zero drawdown, serving as an optimal alternative to short-term deposits.